

Classes of Compact-calibres, Using Spaces of Ultrafilters

In parallel with the investigation of calibres in Chapter 4, we here examine the extent of classes of compact-calibres of spaces. We determine those classes $\mathbf{T}$ of infinite cardinals for which there is a Hausdorff space $X(\mathbf{T})$ such that $X(\mathbf{T})$ does not have compact-calibre α if and only if $\alpha \in \mathbf{T}$; and we show that such a space $X(\mathbf{T})$, when it exists, may be chosen completely regular.

In contrast with its behavior concerning calibre, the cardinal number ω plays no special role with respect to compact-calibre: there are infinite, non-compact, Hausdorff spaces which have compact-calibre ω , and others which do not. Accordingly, the format of Corollary 8.4 is simpler than that of Corollary 4.5 (to which it is similar), and 8.5 and 8.6 are simpler than 4.6 and 4.7, respectively.

The examples we give of spaces with compact-calibre properties prescribed in advance are defined by beginning with the Stone–Čech compactification of a discrete space (which has, like every compact space, compact-calibre α for all $\alpha \geq \omega$) and discarding from it the (non-principal) ultrafilters with various degrees of non-uniformity. The spaces defined in this way certainly do not have Souslin number equal to ω^+ , and in fact we do not know whether the spaces $X(\mathbf{T})$ described above can be chosen to satisfy in addition $S(X(\mathbf{T})) = \omega^+$.

In Corollary 8.12 we note a curiosity: there are completely regular, Hausdorff, non-compact spaces with compact-calibre α for all $\alpha \geq \omega$.

8.1 Theorem. Let α, κ and λ be infinite cardinals, and $X = N_\kappa(\lambda)$. The following statements are equivalent.

- (a) $\kappa \leq \alpha \leq \lambda$ or $\kappa \leq \text{cf}(\alpha) \leq \lambda$;
- (b) X is not pseudo- (α, α) -compact;
- (c) X does not have compact-calibre α .

Proof. (a) $\Rightarrow$ (b). We show first that X is not pseudo- (λ, κ) -

compact. Set $U_\eta = \{\eta\}$ for $\eta < \lambda$ and let $x \in X$. If $x \in \lambda$, then $\{x\}$ is a neighborhood V of x such that

$$|\{\eta < \lambda : V \cap U_\eta \neq \emptyset\}| = 1 < \kappa;$$

and if $x \in X \setminus \lambda$, there is $A \in \mathcal{P}_\kappa^*(\lambda)$ such that $x \in \text{cl}_{\beta(\lambda)} A \subset X$, and then $\text{cl}_{\beta(\lambda)} A$ is an (open-and-closed) neighborhood V of x such that

$$|\{\eta < \lambda : U_\eta \cap V \neq \emptyset\}| = |A| < \kappa.$$

Now if either $\kappa \leq \alpha \leq \lambda$ or $\kappa \leq \text{cf}(\alpha) \leq \lambda$, then (from Theorem 2.1 (c)) either X is not pseudo- (α, α) -compact or X is not pseudo- $(\text{cf}(\alpha), \text{cf}(\alpha))$ -compact. It follows from Theorem 2.3 (c) that X is not pseudo- (α, α) -compact.

(b) $\Rightarrow$ (c). This follows from Theorem 2.1 (f).

(c) $\Rightarrow$ (a). We consider four cases.

Case 1. $\kappa > \lambda$. The space X is then the compact space $\beta(\lambda)$; hence X has compact-calibre α .

Case 2. $\kappa \leq \lambda$ and $\alpha < \kappa$. Let $\{U_\xi : \xi < \alpha\} \subset \mathcal{T}^*(X)$, for $\xi < \alpha$ let $\eta(\xi) \in U_\xi \cap \lambda$, set $A = \{\eta(\xi) : \xi < \alpha\}$ and set

$$K = \text{cl}_{\beta(\alpha)} A.$$

Then K is a compact subspace of X , and

$$\eta(\xi) \in U_\xi \cap K \text{ for } \xi < \alpha.$$

Case 3. $\kappa \leq \lambda$ and $\lambda < \text{cf}(\alpha)$. Let $\{U_\xi : \xi < \alpha\} \subset \mathcal{T}^*(X)$ and set

$$A(\eta) = \{\xi < \alpha : \eta \in U_\xi\} \text{ for } \eta < \lambda.$$

Since $\lambda < \text{cf}(\alpha)$, there is $\bar{\eta} < \lambda$ such that $|A(\bar{\eta})| = \alpha$.

Case 4. $\kappa \leq \lambda$ and $\text{cf}(\alpha) < \kappa$ and $\lambda < \alpha$. It follows from Case 2 (applied to $\text{cf}(\alpha)$ in place of α) that X has compact-calibre $\text{cf}(\alpha)$. Since $\pi w(X) = \lambda < \alpha$, it follows from Theorem 2.3(b) that X has compact-calibre α .

In each of the four cases the space X has compact-calibre α .

The proof is complete.

We see next that in Theorem 2.3(b') the condition $\pi w(X) < \alpha$ cannot be omitted.

8.2 Theorem. Let α be an (infinite) singular cardinal. There is a completely regular, Hausdorff space X such that X has compact-calibre $\text{cf}(\alpha)$, X does not have compact-calibre α , and $\pi w(X) = \alpha$.

Proof. We set $X = N_{(\text{cf}(\alpha))^+}(\alpha)$. It is immediate from Theorem 8.1 that X has the required properties.

We saw in Corollary 3.7(b) that if α is an (infinite) regular cardinal and $\{X_i : i \in I\}$ a set of spaces such that X_i has compact-calibre α for each $i \in I$, then X_I has compact-calibre α . We see next that the analogous statement fails for singular cardinals α . Theorem 8.3 supplements and complements Corollary 3.11(ii), which treats products of spaces with compact-calibre (α, β) with α singular and $\alpha > \beta$.

8.3 Theorem. Let α be an (infinite) singular cardinal. There is a set $\{X_i : i \in I\}$ of completely regular, Hausdorff spaces such that X_i has compact-calibre α for all $i \in I$, and X_I does not have compact-calibre α .

Proof. There is a set $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ of cardinal numbers such that

$$\begin{aligned} \alpha_0 &= \text{cf}(\alpha), \\ \text{cf}(\alpha) &< \alpha_{\sigma'} < \alpha_\sigma < \alpha \text{ for } 0 < \sigma' < \sigma < \text{cf}(\alpha), \text{ and} \\ \sum_{\sigma < \text{cf}(\alpha)} \alpha_\sigma &= \alpha. \end{aligned}$$

We set

$$\begin{aligned} X_\sigma &= N_{(\text{cf}(\alpha))^+}(\alpha_\sigma) \text{ for } \sigma < \text{cf}(\alpha), \text{ and} \\ X &= \prod_{\sigma < \text{cf}(\alpha)} X_\sigma. \end{aligned}$$

That X_σ has compact-calibre α for all $\sigma < \text{cf}(\alpha)$ follows from Theorem 8.1 (with κ and λ replaced by $(\text{cf}(\alpha))^+$ and α_σ , respectively).

We are to show that X does not have compact-calibre α . We show that in fact X does not have compact-calibre $(\alpha, (\text{cf}(\alpha))^+)$.

We set

$$U(\sigma, \eta) = \{x \in X : x_\sigma = \sigma, x_\eta = \eta\} \text{ for } 0 < \sigma < \text{cf}(\alpha), \eta < \alpha_\sigma,$$

and we note that

$$\pi_\sigma[U(\sigma, \eta)] = \{\eta\} \text{ for } 0 < \sigma < \text{cf}(\alpha), \eta < \alpha_\sigma.$$

Let K be a compact subspace of X , and set

$$A(\sigma) = \{\eta < \alpha_\sigma : U(\sigma, \eta) \cap K \neq \emptyset\} \text{ for } 0 < \sigma < \text{cf}(\alpha).$$

For $\eta \in A(\sigma)$ we have $\pi_\sigma[U(\sigma, \eta)] \cap \pi_\sigma[K] \neq \emptyset$, and hence $\eta \in \pi_\sigma[K]$. It follows from Lemma B.10 of the Appendix (with λ, κ and K replaced by α_σ , $(\text{cf}(\alpha))^+$ and $\pi_\sigma[K]$, respectively) that

$$|A(\sigma)| \leq \text{cf}(\alpha) \quad \text{for } 0 < \sigma < \text{cf}(\alpha);$$

hence

$$|\{\langle \sigma, \eta \rangle : U(\sigma, \eta) \cap K \neq \emptyset\}| = \sum \{|A(\sigma)| : 0 < \sigma < \text{cf}(\alpha)\} \leq \text{cf}(\alpha),$$

as required.

The proof is complete.

We give now a consequence of Theorem 8.1, and a theorem which allows us to determine those sets of infinite cardinal numbers which can arise as the class of non-compact-calibres of a space.

8.4 Corollary. Let λ and α be infinite cardinals and $X = N_\lambda(\lambda)$. Then X has compact-calibre α if and only if $\alpha \neq \lambda$ and $\text{cf}(\alpha) \neq \lambda$.

8.5 Theorem. Let S be a non-empty set of infinite cardinal numbers, $X(\lambda) = N_\lambda(\lambda)$ for $\lambda \in S$ and $X(S) = \prod_{\lambda \in S} X(\lambda)$.

For $\alpha \geq \omega$, the following statements are equivalent.

- (a) $X(S)$ does not have compact-calibre α ;
- (b) $\alpha \in S$ or $\text{cf}(\alpha) \in S$.

Proof. (a) $\Rightarrow$ (b). We suppose that $\alpha \notin S$ and $\text{cf}(\alpha) \notin S$. It follows from Theorem 6.1 that $X(\lambda)$ has compact-calibre α for all $\lambda \in S$.

We show that $X(S)$ has compact-calibre α . If α is regular this follows from Corollary 3.7(b). We assume in what follows that α is singular.

Let $\{\alpha_\sigma : \sigma < \text{cf}(\alpha)\}$ be a set of cardinal numbers such that

α_σ is regular for $\sigma < \text{cf}(\alpha)$,

$\text{cf}(\alpha) < \alpha_\sigma < \alpha_\sigma < \alpha$ for $\sigma' < \sigma < \text{cf}(\alpha)$, and

$$\sum_{\sigma < \text{cf}(\alpha)} \alpha_\sigma = \alpha.$$

We claim that if $J \in \mathcal{P}_\omega^*(S)$, then X_J has compact-calibre α and compact-calibre $(\alpha_{\sigma+1}, \alpha_\sigma)$. That X_J has compact-calibre α follows from Theorem 2.2(b); to prove that X_J has compact-calibre $(\alpha_{\sigma+1}, \alpha_\sigma)$ we consider two cases.

Case 1. $\alpha_\sigma \notin J$. Then $X(\lambda)$ has compact-calibre α_σ for all $\lambda \in J$.

It follows that X_J has compact-calibre α_σ and hence compact-calibre $(\alpha_{\sigma+1}, \alpha_\sigma)$.

Case 2. $\alpha_\sigma \in J$. If $J \setminus \{\alpha_\sigma\} = \emptyset$ then X_J , which is $X(\alpha_\sigma)$, has compact-calibre $\alpha_{\sigma+1}$ and hence compact-calibre $(\alpha_{\sigma+1}, \alpha_\sigma)$. If $J \setminus \{\alpha_\sigma\} \neq \emptyset$ then, since $X_{J \setminus \{\alpha_\sigma\}}$ has compact-calibre $(\alpha_{\sigma+1}, \alpha_\sigma)$ (by Case 1) and $X(\alpha_\sigma)$ has compact-calibre $\alpha_{\sigma+1}$, the space X_J , which is $X_{J \setminus \{\alpha_\sigma\}} \times X(\alpha_\sigma)$, has compact-calibre $(\alpha_{\sigma+1}, \alpha_\sigma)$.

The proof of the claim is complete. It follows from Corollary 3.11(b) that $X(S)$ has compact-calibre α , as required.

(b) $\Rightarrow$ (a). From Corollary 8.4 it follows that there is $\lambda \in S$ such that $X(\lambda)$ does not have compact-calibre α . It is then clear that $X(S)$ does not have compact-calibre α .

8.6 Corollary. Let T be a class of infinite cardinal numbers. Then statements (a) and (b) are equivalent.

(a) There is an infinite space X such that X does not have compact-calibre α if and only if $\alpha \in T$.

- (b) (i) If $\text{cf}(\alpha) \in T$ then $\alpha \in T$,
 (ii) $\{\text{cf}(\alpha) : \alpha \in T\}$ is a set, and
 (iii) if κ is a regular cardinal and $\kappa \notin T$,
 then $\{\alpha \in T : \text{cf}(\alpha) = \kappa\}$ is a set.

Proof. (a) $\Rightarrow$ (b). (i) follows from Theorem 2.3(b).

(ii) If α is an infinite cardinal and $\text{cf}(\alpha) > d(X)$, then it follows from Theorems 2.6 and 2.1(d) that X has compact-calibre α . Hence

$$\{\text{cf}(\alpha) : \alpha \in T\} \subset (d(X))^+.$$

(iii) From Theorem 2.3(b') we have

$$\{\alpha \in T : \text{cf}(\alpha) = \kappa\} \subset (\pi w(X))^+ \quad \text{for } \kappa \notin T.$$

(b) $\Rightarrow$ (a). For $\kappa \in \{\text{cf}(\alpha) : \alpha \in T\}$ we set

$$\begin{aligned} S(\kappa) &= \{\kappa\} && \text{if } \kappa \in T, \\ &= \{\alpha \in T : \text{cf}(\alpha) = \kappa\} && \text{if } \kappa \notin T, \end{aligned}$$

we set

$$S = \cup \{S(\kappa) : \kappa \in \{\text{cf}(\alpha) : \alpha \in T\}\},$$

we note that S is a set and $S \subset T$, and we set

$$\begin{aligned} X(\lambda) &= N_\lambda(\lambda) \text{ for } \lambda \in S \text{ and} \\ X &= \prod_{\lambda \in S} X(\lambda). \end{aligned}$$

Suppose next that $\alpha \geq \omega$ and that X does not have compact-calibre α . From Theorem 8.5 it follows that $\alpha \in S$ or $\text{cf}(\alpha) \in S$, and since $S \subset T$ we have $\alpha \in T$ from (i).

Suppose next that $\alpha \in T$. If $\text{cf}(\alpha) \in T$ then

$$\text{cf}(\alpha) \in S(\text{cf}(\alpha)) \subset S,$$

and if $\text{cf}(\alpha) \notin T$ then $\alpha \in S(\text{cf}(\alpha)) \subset S$; it follows from Theorem 8.5 that X does not have calibre α .

8.7 Remark. The space X defined in the proof of the implication (b) $\Rightarrow$ (a) in Corollary 8.6 is a completely regular, Hausdorff space. This shows that if for a class T of infinite cardinal numbers there is a space X such that X does not have compact-calibre α if and only if $\alpha \in T$, then X may be chosen to be a completely regular, Hausdorff space. The spaces X defined above do not, however, satisfy the condition $S(X) = \omega^+$ (cf. Remark 4.8). We do not know whether, given a class T of cardinals satisfying the conditions of Corollary 8.6(b), it is possible to define a completely regular Hausdorff space X satisfying the conditions of Corollary 8.6(a) and in addition the condition $S(X) = \omega^+$.

The remainder of this chapter is preparatory for Theorem 8.11, where we determine the compact-calibres and pseudo-compactness numbers of generalized Σ -products.

8.8 Lemma. Let $\{X_i : i \in I\}$ be a set of Hausdorff spaces, $p \in X_I$, and $\{x(\xi) : \xi < \alpha\}$ a subset of X_I such that

$$\begin{aligned} E_p(x(\xi)) \cap E_p(x(\xi')) &= \emptyset \quad \text{for } \xi' < \xi < \alpha, \text{ and} \\ Y &= \{x(\xi) : \xi < \alpha\} \cup \{p\}. \end{aligned}$$

Then Y is closed in X_I .

Proof. Let $x \in X_I \setminus Y$. Since $x \neq p$ there is $i \in E(x)$ and there is a neighborhood U_i of x_i in X_i such that $p_i \notin U_i$. If $i \notin \bigcup_{\xi < \alpha} E(x(\xi))$, we set $W = \pi_i^{-1}(U_i)$; and if there is $\xi < \alpha$ such that $i \in E(x(\xi))$, then (ξ is unique and) we choose a neighborhood V of x such that $x(\xi) \notin V$ and we set $W = V \cap \pi_i^{-1}(U_i)$. Then W is a neighborhood of x and $W \cap Y = \emptyset$.

8.9 Lemma. Let $\{X_i : i \in I\}$ be a set of σ -compact, Hausdorff spaces, $p_i \in X_i$ for $i \in I$, and $X = \Sigma_\omega(p, X_I)$. Then X is σ -compact.

Proof. For $i \in I$ there is a set $\{X(n, i) : n < \omega\}$ of compact subspaces of X_i such that $X_i = \cup_{n < \omega} X(n, i)$. We assume without loss of generality that

$$\begin{aligned} p_i &\in X(n, i) \text{ for } i \in I, n < \omega, \text{ and} \\ X(n', i) &\subset X(n, i) \text{ for } n' < n < \omega. \end{aligned}$$

For $n < \omega$ we set

$$X(n) = \{x \in X_I : |E(x)| \leq n \text{ and } x \in \prod_{i \in I} X(n, i)\}.$$

Since $\{x \in X_I : |E(x)| \leq n\}$ is closed in X_I and $\prod_{i \in I} X(n, i)$ is compact, the space $X(n)$ is compact for $n < \omega$.

We show $\Sigma_\omega(p, X_I) = \cup_{n < \omega} X(n)$. The inclusion $\supset$ is clear. Let $x \in \Sigma_\omega(p, X_I)$ with $|E(x)| = k < \omega$, let $x_i \in X(n(i), i)$ for $i \in E(x)$, and set

$$\bar{n} = \max \{k, \max \{n(i) : i \in E(x)\}\};$$

then $x \in X(\bar{n})$.

8.10 Theorem. Let α be an (infinite) singular cardinal with $\text{cf}(\alpha) = \omega$ and let $\kappa > \omega, \lambda \geq \omega$, and $X = \Sigma_\kappa \{0, 1\}^\lambda$. Then X has compact-calibre α .

Proof. The space $\Sigma_{\omega^+} \{0, 1\}^\lambda$ is dense in X ; it follows from Theorem 2.5(a)(ii) that it is sufficient to prove the theorem in the case $\kappa = \omega^+$. In what follows we assume that $\kappa = \omega^+$ and we consider three cases.

Case 1. $\lambda < \alpha$. It follows from Theorem 3.8 that X has compact-calibre ω . Since

$$\pi w(X) \leq w(X) \leq w(\{0, 1\}^\lambda) = \lambda < \alpha,$$

the space X has compact-calibre α by Theorem 2.3(b').

Case 2. $\lambda = \alpha$. Let $\{U^\xi : \xi < \alpha\}$ be a set of non-empty basic subsets of $\{0, 1\}^\alpha$, and let $\{\alpha_n : n < \omega\}$ be a set such that

$$\begin{aligned} \alpha_n &\text{ is a regular cardinal for } n < \omega, \\ \omega &< \alpha_{n'} < \alpha_n < \alpha \quad \text{for } n' < n < \omega, \text{ and} \\ \sum_{n < \omega} \alpha_n &= \alpha. \end{aligned}$$

It follows from Lemma 1.9 and Theorem A.7 of the Appendix

(with $\kappa = \omega$) that there are $\{A_n : n < \omega\}$ and $\{J_n : n < \omega\}$ such that

$$\begin{aligned} A_n &\in [\alpha]^{\alpha_n} \quad \text{for } n < \omega, \\ A_{n'} \cap A_n &= \emptyset \quad \text{for } n' < n < \omega, \text{ and} \\ R(U^{\xi}) \cap R(U^{\xi'}) &= J_n \quad \text{for } \xi, \xi' \in A_n, \xi' \neq \xi \\ &\subset J_n \quad \text{for } \xi' \in A_{n'}, \xi \in A_n, n' < n < \omega. \end{aligned}$$

We set

$$A = \bigcup_{n < \omega} A_n, \quad J = \bigcup_{n < \omega} J_n, \quad I = \alpha \setminus J, \text{ and}$$

$$V^{\xi} = \pi_I[U^{\xi}] \quad \text{for } \xi \in A,$$

and we note that

$$\begin{aligned} |A| &= \alpha, |J| \leq \omega, \text{ and} \\ R(V^{\xi'}) \cap R(V^{\xi}) &= \emptyset \quad \text{for } \xi', \xi \in A, \xi' \neq \xi. \end{aligned}$$

For $\xi < \alpha$ we define $x(\xi)$ by

$$\begin{aligned} x(\xi)_\eta &\in V_\eta^{\xi} \quad \text{if } \eta \in R(V^{\xi}) \\ &= 0 \quad \text{if } \eta \in I \setminus R(V^{\xi}). \end{aligned}$$

We note that if $\xi', \xi \in A$ and $\xi' \neq \xi$, then

$$E(x(\xi')) \cap E(x(\xi)) \subset R(V^{\xi'}) \cap R(V^{\xi}) = \emptyset;$$

we note further that if $\xi \in A$, then

$$|E(x(\xi))| \leq |R(V^{\xi})| < \omega$$

and hence $x(\xi) \in \Sigma_\omega \{0, 1\}^I$. We define $p \in \{0, 1\}^I$ by

$$p_\eta = 0 \quad \text{for } \eta \in I,$$

we set $K_I = \{x(\xi) : \xi \in A\} \cup \{p\}$, and we note from Lemma 8.8 that K_I is closed in $\{0, 1\}^I$ and hence compact.

We set $K = \{0, 1\}^J \times K_I$. Then K is compact, and since $|J| \leq \omega$ and $K_I \subset \Sigma_\omega \{0, 1\}^I$, we have $K \subset \Sigma_{\omega+} \{0, 1\}^\alpha$. Finally, for $\xi \in A$ there is $y(\xi)$ such that

$$\begin{aligned} y(\xi)_J &\in \pi_J[U^{\xi}], \text{ and} \\ y(\xi)_I &= x(\xi); \end{aligned}$$

it is then clear that $y(\xi) \in U^{\xi} \cap K$, as required.

Case 3. $\lambda > \alpha$. Let $\{U^{\xi} : \xi < \alpha\}$ be a set of non-empty basic subsets of $\{0, 1\}^\lambda$, and set $F = \bigcup_{\xi < \alpha} R(U^{\xi})$. Since $|F| \leq \alpha$ it follows from

Cases 1 and 2 that there are $A \in [\alpha]^\alpha$ and compact $K_F \subset \Sigma_{\omega^+} \{0, 1\}^F$ such that $\pi_F[U^\xi] \cap K_F \neq \emptyset$ whenever $\xi \in A$. We define

$$p = \langle p_\eta : \eta \in \alpha \setminus F \rangle \in \{0, 1\}^{\alpha \setminus F}$$

by

$$p_\eta = 0 \quad \text{for } \eta \in \alpha \setminus F$$

and we set $K = K_F \times \{p\}$. Then K is a compact subspace of X , and $U^\xi \cap K \neq \emptyset$ whenever $\xi \in A$, as required.

The proof is complete.

8.11 Theorem. Let α, κ and λ be infinite cardinals and $X = \Sigma_\kappa \{0, 1\}^\lambda$. The following statements are equivalent.

- (a) $\text{cf}(\alpha) = \omega$ and $\kappa = \omega$;
- (b) X is not pseudo- (α, α) -compact;
- (c) X does not have compact-calibre α .

Proof. (a) $\Rightarrow$ (b). We have $\omega \subset \lambda$. We set

$$U_k = \{x \in X : x_n = 1 \text{ for } n \leq k\} \text{ for } k < \omega,$$

we note that $\{U_k : k < \omega\} \subset \mathcal{T}^*(X)$, and we show that the (indexed) family $\{U_k : k < \omega\}$ is locally finite in X . For $x \in X$ we choose $\bar{n}$ such that $x_{\bar{n}} = 0$, we set $U = \pi_{\bar{n}}^{-1}(\{0\})$, and we note that

$$\{k < \omega : U \cap U_k \neq \emptyset\} = \bar{n}.$$

It follows that X is not pseudo- (ω, ω) -compact. Since $\text{cf}(\alpha) = \omega$, it follows from Theorem 2.3(c) that X is not pseudo- (α, α) -compact.

(b) $\Rightarrow$ (c). This follows from Theorem 2.1(f).

(c) $\Rightarrow$ (a). We show that (c) fails if (a) fails.

We consider three cases.

Case 1. $\text{cf}(\alpha) > \omega$. From Lemma 8.9 there is a set $\{K_n : n < \omega\}$ of compact spaces such that

$$\Sigma_\omega \{0, 1\}^\lambda = \bigcup_{n < \omega} K_n.$$

Let $\{U^\xi : \xi < \alpha\} \subset \mathcal{T}^*(X)$, and set

$$A_n = \{\xi < \alpha : U^\xi \cap K_n \neq \emptyset\} \quad \text{for } n < \omega.$$

Since $\Sigma_\omega \{0, 1\}^\lambda$ is dense in X , we have $\bigcup_{n < \omega} A_n = \alpha$, and since $\text{cf}(\alpha) > \omega$, there is $\bar{n} < \omega$ such that $|A_{\bar{n}}| = \alpha$, as required.

Case 2. $\alpha = \omega$ and $\kappa > \omega$. From Theorems 3.8 and 4.1(a) it follows that $\Sigma_{\omega^+} \{0, 1\}^\lambda$ has compact-calibre α . Since $\kappa > \omega$ the space $\Sigma_{\omega^+} \{0, 1\}^\lambda$ is dense in X , and hence X has compact-calibre ω by Theorem 2.5(a) (ii).

Case 3. $\alpha > \omega$, $\text{cf}(\alpha) = \omega$, and $\kappa > \omega$. It follows from Theorem 8.10 that X has compact-calibre α .

The proof is complete.

We note the following surprising consequence of Theorem 8.11.

8.12. *Corollary.* There are completely regular, Hausdorff, non-compact spaces X with compact-calibre α for all $\alpha \geq \omega$.

Proof. We set $X = \Sigma_\kappa \{0, 1\}^\lambda$, with $\kappa > \omega$ (to ensure that X has compact-calibre α) and with $\lambda \geq \kappa$ (to ensure that X is not compact).

For a specific example, we set $X = \Sigma_{\omega^+} \{0, 1\}^{(\omega^+)}$.

We note that from Theorem 2.1(f) it follows that the (non-compact) spaces X of Corollary 8.12 are pseudo- (α, α) -compact for all $\alpha \geq \omega$.

Notes for Chapter 8

We announced most of the results of this chapter in the abstract of Comfort & Negreponis (1978b); see also Comfort (1980).